



Technical Report

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Course & Semester:

DX 703 O1: Advanced Machine Learning and AI

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Dataset Used:

HuffPost News Category

B.1: Introduction and Problem Definition

B.1.1: Problem Description and Motivation

This project focuses on a multi-class classification problem using the text-based HuffPost News Category dataset. The primary goal of this study is to build a classification model that automatically assigns new articles to their appropriate category based on their headline and short description. As digital news platforms publish numerous articles on a wide range of topics daily, manually organizing them into categories can be inefficient and inconsistent. Without automated classification systems, users may experience difficulty in finding relevant articles, while news platforms are unable to effectively provide tailored content to their users. This problem is crucial to address, as accurate and reliable classification can improve content organization, search optimization, and offer personalized recommendations. Furthermore, it can enhance the user experience by enabling quick article filtering, reducing search time, and making the platform more accessible.

B.1.2: Project Goals in Business and Data Science

The business goal for this project is to automate the process of categorizing news articles to ensure that the content is efficiently organized and distributed to users. This includes faster labeling and tagging of articles, better search and filtering capabilities, and improved overall user experience. The data science goal is to develop a multi-class classification model that predicts the correct category of a new article based on its text input. The model will use the article's headline and short description as input, along with a [SEP] token for separating text segments. It should accurately result in one of the 41 predefined categories. Also, the model should be able to handle challenges such as class imbalance, data noise, and any inconsistencies in the text data to ensure strong performance across classes. Evaluation metrics such as per-class precision and

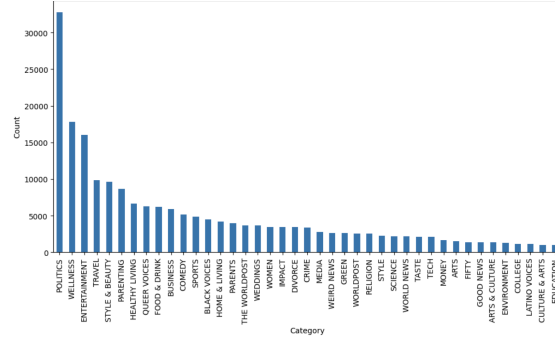
recall will be used to better understand how the model performs on individual categories, with an emphasis on macro F1 to ensure that all classes are treated equally.

B.1.3: Dataset Description

The dataset used for this analysis is the HuffPost News Category dataset, which was accessed from the Hugging Face platform. It is a text-based dataset that contains 200,853 news samples, which are comprised of article headlines and their short descriptions. Also, the dataset consists of 41 different categories that describe news topics, such as POLITICS, ENTERTAINMENT, TRAVEL, and SPORTS. The input modality of the dataset is text-based data. A [SEP] token is used to separate the article headline and short description within a single input sequence. The key features used include the headlines, short descriptions, and the 41 categories. One challenge in the data is the class imbalance, with the largest class being POLITICS, containing 32,739 samples, while categories like CULTURE & ARTS and EDUCATION have 1,030 and 1,004, respectively. This imbalance can lead to biases in the model toward more frequent classes, resulting in inaccurate and unreliable predictions for minority classes. This can reduce overall model fairness and performance across all classes. Another significant challenge is overlapping or ambiguous categories, such as POLITICS versus WORLD NEWS and WELLNESS versus HEALTHY LIVING. This can cause the model to struggle in distinguishing between these categories, leading to misclassifications and potentially decreasing predictive performance. Other issues observed included 488 duplicate entries, variations in text length, such as 473 short descriptions of 3 words or fewer, and 5 missing texts, as well as headlines being vague or lacking context. All of these can introduce noise during model training, potentially impacting generalizability and reducing classification accuracy. Below are two EDA figures, showing class frequency count (left) and class distribution (right).

Top Categories	Count	Bottom Categories	Count
POLITICS	32739	ENVIRONMENT	1323
WELLNESS	17827	COLLEGE	1144
ENTERTAINMENT	16058	LATINO VOICES	1129
TRAVEL	9887	CULTURE & ARTS	1030
STYLE&BEAUTY	9649	EDUCATION	1004

Caption. Class frequency distribution table showing the five largest and five smallest categories by sample count.



Caption. Class distribution of the 41 categories, showing severe class imbalance in the data.

B.2: Methodology

B.2.1: Preprocessing Pipeline

The preprocessing pipeline focused on preparing the text data for multi-class text classification. The dataset was cleaned by removing duplicate entries, filtering out empty records, and eliminating very short text samples lacking meaningful information. The headline and short description fields were combined into a single input using a “[SEP]” token, allowing the model to capture both concise and contextual information. Also, the text was standardized by converting it to lowercase and removing extra white space to ensure consistency. Labels were encoded into integers using a label encoder to improve interpretability, and class imbalance was addressed by computing class weights based on training data distribution. For the final model, text inputs were processed using the DistilBERT preprocessor for tokenization and contextualized token representations. This approach also handles rare or unseen words through subword tokenization. A maximum sequence length of 128 tokens was used, with padding and truncation applied to maintain consistent input dimensions.

B.2.2: Model Architecture and Justification

The final model selected for this project is a pretrained DistilBERT backbone (distil_bert_en_uncased) that generates contextual embeddings, followed by a classification layer with softmax activation to produce probabilities across the 41 classes. The selection of DistilBERT was based on results from Milestone 2, where it outperformed both the baseline embedding-based model and the custom BiGRU model. Further improvements were achieved through systematic hyperparameter tuning in the final notebook. The final configuration was trained using early stopping based on validation accuracy, ensuring that the model retained the best-performing weights and avoided overfitting. This approach resulted in strong performance and improved generalization, making DistilBERT the most effective model for this task.

B.2.3: Training Strategy

For the final model, the loss function used was sparse categorical cross-entropy, which is suitable for multi-class classification with a large number of classes. Since the HuffPost dataset contains 41 categories, this loss function compares predicted probabilities with true labels, enabling the model to learn meaningful distinctions and improve accuracy. The optimizer used was AdamW, which is standard for fine-tuning transformer models like DistilBERT. AdamW applies weight decay, improving generalization and reducing overfitting while maintaining stable training. The final configuration used a learning rate of 1e-5 and weight decay of 0.01, chosen to improve stability and reduce overfitting observed in earlier experiments. The model used a batch size of 32, balancing training efficiency and stable updates, and was trained for 10 epochs with early stopping based on validation performance. A maximum sequence length of 128 tokens was used to better capture context from headlines and descriptions. The dropout rate remained at the default 0.1 to help prevent overfitting. The callback used was EarlyStopping (monitor="val_accuracy", patience=3, restore_best_weights=True, mode="max"). Validation

accuracy was monitored instead of validation loss because earlier experiments showed validation loss diverging while accuracy improved. This allowed training to continue as long as performance improved, while patience ensured stopping after three epochs without improvement. Restoring the best weights ensured the model retained the most effective parameters.

B.2.4: Performance Validation

The dataset was split into training, validation, and test sets with an 80/10/10 stratified split. The validation set was used during training to monitor and evaluate model performance after each epoch and to guide hyperparameter tuning, while the test set was kept separate and used only for final evaluation to measure model performance. A stratified split was applied based on the target labels to ensure that all 41 categorical classes were preserved and equally represented in all sets. To further address class imbalance, class weights were set to “balanced” to ensure that underrepresented classes were contributing to the loss function more during training and would reduce bias toward majority classes. To handle data leakage, a stratified split was implemented before pre-processing and tokenization, and the test set was isolated. This helped avoid information from the test set influencing model training.

B.3: Results and Evaluation

B.3.1: Quantitative Results

The final model was evaluated using **accuracy, macro-F1 score, and weighted F1-score** to capture overall and class-balanced performance. We present **training and validation accuracy and loss curves (left)** to analyze learning behavior, along with **data-specific metrics for the final model test performance (right)** to evaluate final model performance. Also, a **confusion matrix for the test set** is provided in the Appendix to illustrate prediction errors and misclassifications across all 41 categories.



Caption. Training and validation accuracy (left) and loss (right) over epochs for the final model.

Final Model Test Performance

Test Accuracy	0.6652
Macro-F1	0.5927
Weighted-F1	0.6741

Caption. Final model performance, showing accuracy, macro-f1, and weighted-f1 scores.

B.3.2: Interpretation of Results

The results show clear variation in performance across categories. Classes such as STYLE & BEAUTY, TRAVEL, and WEDDINGS are easier for the model to classify, as they contain more distinct and consistent vocabulary. In contrast, categories such as GOOD NEWS, WOMEN, and IMPACT are more difficult, likely due to their broader and more ambiguous nature. The confusion matrix shows that most predictions lie along the diagonal, indicating generally strong classification performance. However, there are noticeable patterns of confusion between semantically similar categories. For example, PARENTING and PARENTS, as well as WELLNESS and HEALTHY LIVING, are frequently misclassified. Additionally, the category POLITICS shows a wide spread of predictions across related categories such as WORLD NEWS, MEDIA, and BUSINESS, reflecting overlap in content and vocabulary. These patterns suggest that the model struggles when categories share similar themes. There is also evidence of mild overfitting. Training accuracy continues to increase over epochs, while validation accuracy stabilizes early and validation loss begins to rise. This indicates that the model is learning patterns specific to the training data but does not fully generalize to unseen data. The use of early stopping helps reduce this effect by selecting the best-performing model. Finally, the impact of

class imbalance is clearly visible. Larger classes, such as POLITICS, dominate predictions and show stronger performance, whereas smaller classes, such as EDUCATION and LATINO VOICES, are underrepresented and perform worse. This imbalance makes it more difficult for the model to learn meaningful patterns for underrepresented categories, even with the use of class weights.

B.3.3: Summary of Final Model Performance

The **final DistilBERT model** represents the **best solution** for this task due to its ability to capture contextual meaning in text and outperform simpler baseline and custom models. By leveraging a pretrained transformer and fine-tuning it on the dataset, the model achieves strong overall performance with a test accuracy of 0.6652 and a macro F1-score of 0.5927, indicating balanced performance across classes. These results also highlight the difficulty of the task. With 41 categories, significant class imbalance, and overlapping label definitions, perfect classification is challenging. While the model performs well on clearly defined categories with distinct vocabulary, it struggles with broader or semantically similar classes. A key strength of the model is its ability to generalize across diverse text inputs, especially for well-defined categories. However, its main weakness lies in handling ambiguous and underrepresented classes, where performance is lower due to limited data and overlapping language patterns. Overall, the results demonstrate that while DistilBERT is effective for this problem, further improvements could focus on better handling class imbalance and label ambiguity.

B.4: Discussion and Reflection

B.4.1: Limitations and Sources of Error

There were various limitations present in the dataset and final model. The HuffPost News Category dataset has a significant class imbalance across its 41 categories. Some categories, such

as POLITICS, have thousands of samples, while others, like CULTURE & ARTS and EDUCATION, have very few. This severe imbalance makes it difficult for the model to learn meaningful patterns for underrepresented classes, even with class weights, leading to lower prediction accuracy and poor performance on minority classes. Also, multiple categories were either overlapping or ambiguous, including WELLNESS vs HEALTHY and POLITICS vs WORLD NEWS, which could cause noise and lead to misclassifications. Also, the dataset contained duplicate text entries, which could introduce redundancy and potential bias towards repeated patterns. Lastly, some text entries were either very short or missing short descriptions, which can reduce the amount of contextual information available, thus limiting the model's ability to distinguish between categories.

B.4.2: Potential Improvements

If more time were given for this project, multiple improvements could be made to enhance model performance and robustness. For example, to further address class imbalance, one improvement is focal loss, which reduces the weights of classes that have been effectively learned and classified, while placing more emphasis on classes that are more difficult and misclassified. This improvement helps the model focus more on the minority classes, thereby improving performance on underrepresented categories and reducing bias towards the majority classes. Another improvement that could be made is implementing a more expressive architecture, such as RoBERTa or even DeBERTa-v3. These transformer models use stronger contextual representations and are more effective at identifying the most important words for classification than DistilBERT. This allows them to better capture semantic differences between similar categories. A third improvement that could be included is the use of additional data cleaning and balancing techniques. This would include more thorough refinement of overlapping

and ambiguous categories, such as combining similar categories and clearly defining their boundaries to reduce confusion and misclassifications. Also, this would involve a more systematic approach to handling very short descriptions or entries that are near-duplicates to ensure that there is sufficient contextual information for model learning and reliable classification. Overall, these improvements can help strengthen model performance and generalization to unseen data.

B.4.3: Reflection On What We Learned

One of the most significant factors in the model's performance was the dataset's quality and structure. Despite using a strong transformer model such as DistilBERT and various tuning strategies, performance improvements were marginal, suggesting the model was limited by the data itself. Specifically, the severe class imbalance and overlapping categories had the most impact on the results. Categories that were distinct and clearly defined performed significantly better than those that were ambiguous or semantically similar. In this project, hyperparameter tuning helped make small improvements to the model's performance. This showed that optimization helped refine performance but did not significantly improve predictive performance or yield large gains in macro F1 score and accuracy. Tuning, such as lowering the learning rate, switching to the AdamW optimizer, and increasing sequence length, slightly increased accuracy and macro F1 score. However, the gap between training and validation performance indicates that these tuning approaches were insufficient to significantly improve generalization. Overall, tuning can be beneficial for stabilizing and enhancing models like DistilBERT, but model performance can be constrained by data quality and structure rather than by optimization choices.

If the project were to be repeated, what would be done differently is to further address class imbalance by resampling to better represent minority classes and by using data

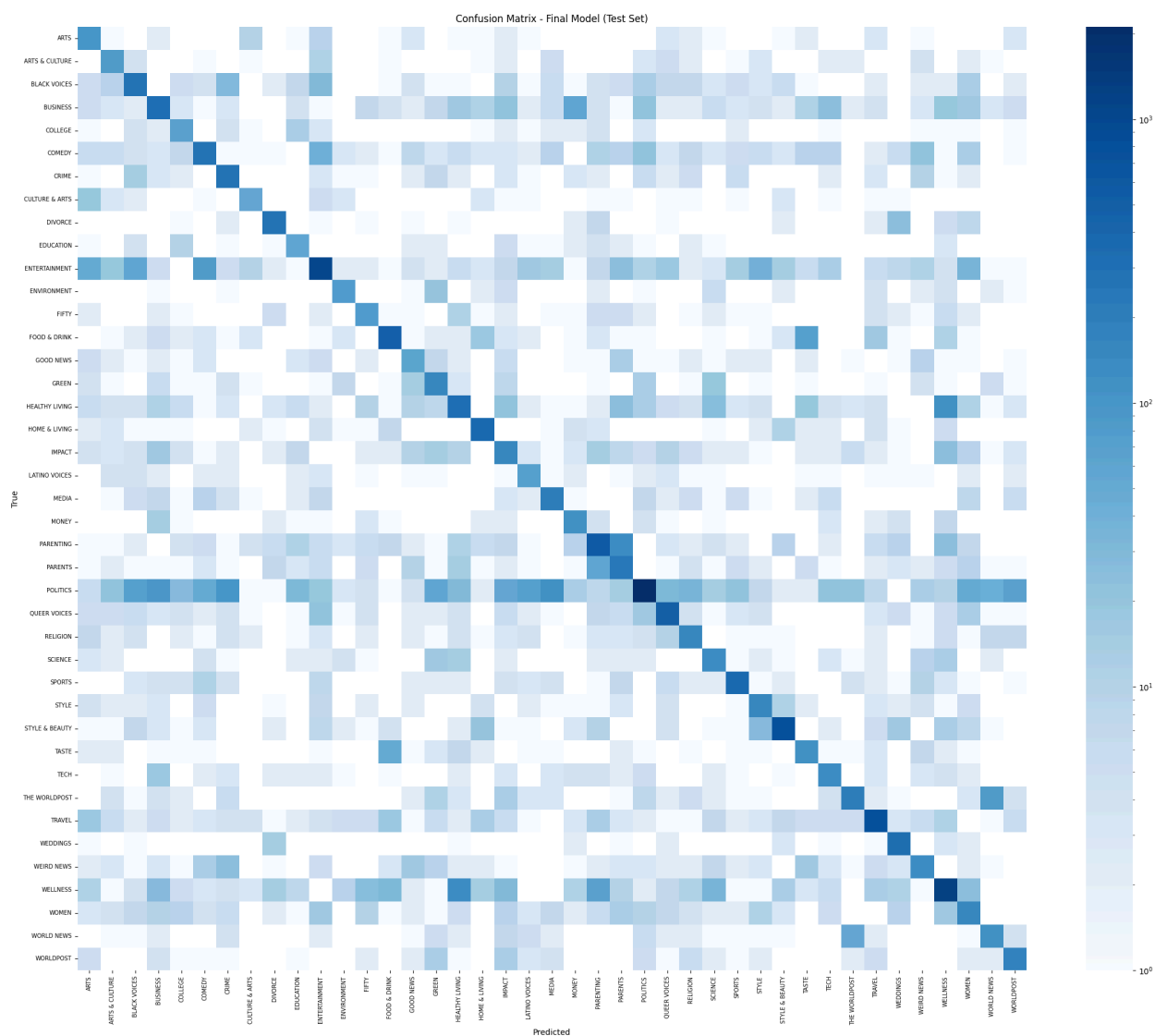
augmentation to increase the size and diversity of smaller classes. This can help improve model performance, especially in the macro F1 score. Also, merging similar categories and hierarchical classification would be implemented to reduce ambiguity and make the classification task easier to learn. A key insight is that in multi-class classification problems, such as in this project, model performance is strongly influenced by dataset structure and characteristics rather than model complexity or optimization approaches. This indicates the importance of data preprocessing and how well class imbalance and class ambiguity are addressed.

B.5: AI Use Disclosure

To complete this final project, we referenced our coding notebooks and homework to perform EDA and build our final DistilBERT model. We referenced the video notebooks from 703 to explain how we will prepare and improve the data to address various challenges we have identified. Additionally, we used our codes and analysis from homework 9 to help build a baseline DistilBERT model and to make further improvements. We did use this one article for understanding and building the text vectorization layer:

https://keras.io/api/layers/preprocessing_layers/text/text_vectorization/. This link helped break down all the components, which was crucial given that we used Keras to help build our models across the final project. This article has an in-depth example of how each step is built, what each feature means, and how it is relevant to the computation. Also, we used ChatGPT to evaluate our paper against the instructions and rubric of the project to determine if improvements needed to be made. This allowed us to ensure we met the criteria for each section without being too repetitive. Here is the link: <https://chatgpt.com/share/69ee86d3-e1c0-83e8-bd83-e361f4395e8b>.

Appendix



Caption. Final model confusion matrix for the test set, showing the distribution of predicted versus true labels across all 41 categories.